



Cloud / Central Server



Global
Aggregator



Model Update
Control



TEE-Protected
Server Enclave



Secure
Aggregation



Trust
Audit



Key
Management



Access
Control

5 Trust Audit & Secure Aggregation

6 Global Model Update

4 Secure WAN /
TLS Channel



Edge Gateway Layer

Edge Gateway (Representative)



Client
Manager



Local
Model Pool



Privacy-Preserving
Communication



Additional Edge Gateways

7 Model Return to Edge
and Clients

3 Encrypted
Local Update

2 Attestation /
Trust Evidence



Trusted Client Device Layer

Trusted Client Device (Representative)



TEE /
TMAA
Enclave



Secure
Local
Training



Private
Dataset



Local
Model
State



Attestation
Quote



Encrypted
Update



Additional Client Devices

1 Local Data & Training